Does Money Buy Wins? A Machine Learning Approach to Understanding Team Performance

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Does Payroll Buy Wins? A Machine Learning Approach to Understanding Team Performance

In recent years, Major League Baseball (MLB) has become a proving ground for advanced analytics, reshaping how teams evaluate talent and allocate resources. The game used to solely judge players based on their batting average, and for pitchers, the Earned Run Average (ERA). The analytics boom in baseball has since shown us that these statistics are insufficient and lack in accurately reflecting a player's true value. Consequently, more game aspects have been taken into account when team management, experts, and fans predict a team's performance. Sabermetrics, such as Wins Above Replacement (WAR) and expected statistics (expected results versus actual outcomes based on league probabilities), are now considered the most influential statistics. Meanwhile, the ideology of a money-performance association still exists in people's minds, considering the huge boom in payroll in MLB, team managers want to ensure they get their money's worth when investing in a player.

Teams in the league adopted various data analytics approaches, which provided different levels of financial availability. Some franchises achieve high WAR totals and playoff berths with modest budgets, while others spend heavily yet fail to deliver consistent results. The separation of budgets in the MLB has raised important questions about efficiency, sustainability, and competitive balance within the league. As a result, field experts and researchers have been trying to determine aspects and indicators that represent a team's performance, and understand the relationship between such aspects and the performance. Miceli and Huber (2009) examined teams' performance from 1985 to 2001 using payroll information and winning percentage, and found a positive and significant relationship between higher pay and more wins within each season. WAR was then further introduced and has been used ever since to demonstrate the importance and contribution to the field of baseball in evaluating players' performance (Baumer

et al., 2015). Therefore, the current study plans to follow the same direction to further investigate the relationship between payroll, WAR, and team performance using more recent financial and ranking data. Linear regression was used to determine whether payroll and WAR are determinants of winning percentage, both individually and collectively. Additionally, an investigation using random forest (RF) was conducted to determine if either of these aspects is useful in predicting a team's divisional title-winning result, appearance in the season's playoffs, World Series appearance, and World Series title-winning result. It is hypothesized that (1) WAR and payroll are good at predicting winning rates both individually and collectively, and (2) that WAR is indicative of a team's chance to win the division title and the World Series, but not for appearances in playoffs or World Series.

Method

Measures

Team Payroll Data

Team payroll data is publicly available on Spotrac (spotrac.com), a platform that tracks team payrolls, contracts, and salary distributions across Major League Baseball. Financial data was extracted between 2015 to 2025, while the shortened 2020 season was excluded due to the global pandemic.

To ensure the data is clean and can be used directly for analysis, the dollar signs and thousands separators were removed.

Team Performance Data

All team performance data is sourced from FanGraphs (fangraphs.com), a widely respected platform that provides comprehensive player and team metrics, including Wins Above Replacement (WAR). Teams' performance data, including their individual winning percentage,

WAR indicator, and playoff and World Series participation, were extracted from 2015 to 2025 without the 2020 season for the same reason mentioned above. The winning percentage is normalized to ensure it is below 1.0, and the other information is rearranged into a dichotomous format, with 1 representing True (i.e., a team won the division title, appeared in the playoffs, appeared in the World Series, and/or won the World Series) and 0 representing False.

Software

Google Colaboratory was used for data cleaning, analysis, and visualization (Google, 2025). Python 3.12.12 is the default programming language in the system and is used throughout the project for the previously mentioned processes.

Results

Descriptive statistics were run to obtain a general understanding of each team's expenses and WAR indicator across the years. Tables 1 and 2 in Appendix A show the mean and standard deviation as well as 25%, 50% and 75% percentiles of each variable. Figure 1 also illustrates the average payroll in relation to the average WAR indicator for each team over a 10-year period.

Correlation analyses were performed, and WAR was found to be strongly and significantly related to winning percentage ($r = 0.8501$, $p < .01$), while a moderately significant relationship was found between total payroll and winning percentage ($r = 0.349$, $p < .01$).

Additionally, an OLS (Ordinary Least Squares) linear regression was conducted to determine whether the independent variables (payroll and WAR) predict the dependent variable (winning percentage). The first run of the analyses used both payroll and WAR indicator and revealed that both predictors explained 72.2% of the variance ($R^2 = .723$, $F(2,297) = 388.3$, $p < .01$). It was found that WAR significantly predicted winning percentage ($\beta = .0056$, $p < .001$),

while total payroll does not (β = 4.018e-11, p = .382). The model has a value of 12.9 for the sum of squared error (SSE).

To further understand the role of WAR, a WAR-only model was conducted, and it was found that WAR could account for 72.2% of the variance (R^2 = .723, $F(1,298)$ = 776.5, $p < .01$) and is a strong positive predictor of winning percentage (β = .0056, $p < .001$). Similar to the previous analysis, this model also had a value of 0.52 for SSE.

A third run of the model with total payroll was also performed. Such a model explained approximately 11.9% of the variance in winning percentage (R^2 = .122, $F(1,298)$ = 41.24, $p <$ .01). The result was extremely small yet statistically significant (β = 4.85e-10, $p < .001$), showing a negligible practical effect. The model also has an SSE of 1.64, which is also considered high.

Team rankings were produced by ranking the predicted winning percentage calculated using the OLS Regression result obtained above. The full list of rankings is presented in the attached file. A t-test was further performed to examine if there is a significant difference between the predicted and actual rankings. It was found that predicted rankings did not differ significantly from the actual rankings, $t(1)$ = -.346, p = 0.729. Predicted Wins against Actual Wins are shown in Figure 2 in Appendix A, and the predicted linear relationship is further compared to a y=x line.

The random forest algorithm demonstrated that total payroll and WAR collectively achieve 78.7% accuracy in predicting division winners, 80.0% accuracy in predicting playoff appearances, 92.0% accuracy in predicting World Series appearances, and 96.0% accuracy in predicting World Series wins. WAR has always been considered a stronger factor in making predictions than payroll; however, in the random forest analysis, the difference between the two

predictors is not significant. The feature importance of WAR was 0.624 when predicting the division winner, 0.689 when predicting playoff appearances, 0.587 when predicting a World Series appearance, and 0.611 when predicting the World Series winner.

When using WAR solely to predict the four dichotomous variables, the model achieved 80.0% accuracy in predicting the division winner, 77.3% accuracy in predicting a playoff appearance, 89.3% accuracy in predicting a World Series appearance, and 92.0% accuracy in predicting the World Series Winner.

Discussion

Payroll and the WAR indicator provide the team management, experts, and fans with a useful tool to monitor and predict their team's performance. The current study followed a similar approach to that of Miceli and Huber (2009), who had previously investigated the roles that total payroll and WAR played in producing such predictions. Initial correlations showed promising results, with a significantly moderate to strong relationship between both variables and the winning percentage. Since correlations do not show predictions, further analyses were conducted to determine the role of payroll in addition to understanding the role of WAR in predicting winning percentage.

The positive and significant linear regression result provided an overview of the predictive relationship between the WAR indicator and total payroll, as well as the winning percentage. The significant contribution of the WAR indicator is promising as it demonstrates the importance of such an indicator in team valuation and performance prediction. However, the combination of WAR and total payroll only showed a relatively low total prediction error, with an SSE of 12.9. However, considering the large sample size, the error is acceptable.

Due to the significant difference between the two variables, two additional tests were also conducted. When considering only WAR as the predictor, a strong positive factor was found to predict the winning percentage. On the other hand, the analysis with only total payroll showed a weak relationship, explained little variance, and displayed worse predictive performance relative to WAR. Altogether, the findings showed that WAR captures the true on-field performance contributions of players more effectively than payroll does, and payroll is a far weaker predictor of a team's win-loss record. The results align well with Baumer and colleagues' research (2015) but contradict those from Miceli and Huber (2009). They are certainly surprising, and yet provide more insight into the direction future studies could take.

Thanks to the significant regression model, a close prediction of the division standing was shown. When comparing each predicted ranking with the actual ranking using the t-test, insignificant results were obtained, indicating that there is no significant difference between the predicted and actual rankings. Additionally, the linear function $y = 0.72x + 22.48$, which represents the data, is close to the reference line $y = x$.

Last but not least, the Random Forest algorithm provided insight into the non-linear relationship between the variables, and it was found that the combination of total payroll and WAR has high accuracy in predicting division winners, playoff appearances, World Series appearances, and World Series wins. Despite a weak involvement in predicting winning percentage, payroll appeared to be relatively equally important in predicting these four areas and produced an overall higher accuracy in all four areas, even though WAR remained more important and carried more weight in the analysis. When running the same algorithm for the second time with only the WAR indicator, it was surprising to see lower accuracy in predicting playoff appearances, World Series appearances, and World Series wins. This could be the reason

that a team with less famous and highly paid players could still be competitive in the title challenge process; however, due to a lack of experience playing in bigger games, they might not perform as well as other highly paid players who have more experience in such playoff and World Series games (Demiralp et al., 2012).

Overall, hypotheses (1) and 2) were both partially supported. It was found that WAR is the only positive and strong indicator in predicting the winning rate. While the combination of WAR and total payroll showed no difference in prediction accuracy, the model with WAR solely is preferred, considering the additional resources and computational power needed for including more variables in model construction and calculation. Regarding whether a team won the division title, the World Series, or appeared in the playoffs or the World Series, WAR has a strong predictive power in all of these outcomes. More interestingly, despite payroll playing an extremely weak role in predicting team performance, it seemed to be equally important as WAR in predicting these four areas.

Limitations and Future Directions

With a significantly large dataset, multicollinearity and confounding variables are not adequately addressed. While it is clear that payroll has small effects in predicting winning percentage, it is hard to tell if the effect in every aspect of analysis is insignificant, provided that the overly strong effect of WAR could easily suppress payroll's effects. Therefore, future studies should continue to investigate the effect of payroll on a team's performance, not only in terms of winning percentage, but also in other aspects, as payroll significantly influences roster construction and is considered a crucial factor in baseball analytics.

In addition, the current study focused solely on the team level and intentionally mitigated the effects of individuals. Demiralp and colleagues (2012) found that the experience players

acquire as they age is beneficial to teams; therefore, two teams could have a similar winning percentage and performance with two drastically different roster-building strategies. In the future, it would be an interesting area for researchers to examine how much effect explicitly more seasoned and experienced players could bring to a team, and if team performance differs when comparing various team profiles.

Moreover, the shortened season 2020 was not included in the current analysis due to the global pandemic. However, due to the significant economic challenges that followed the pandemic, it would be an interesting area for future research to examine whether a team's profile (i.e., WAR valuation and budget availability) is drastically different post-pandemic compared to pre-pandemic, and if such a different profile results in differences in team performance. Altogether, the current study, which analyzes the performance of baseball teams in MLB, is effective and provides valuable insights into sports analytics. It is anticipated that further studies will be conducted in the near future, building on the evidence from both prior and current research.

References

Baumer, B. S., Jensen, S. T., & Matthews, G. J. (2015). OpenWAR: An open source system for evaluating overall player performance in major league baseball. Journal of Quantitative Analysis in Sports, 11(2), 69-84. https://www.degruyterbrill.com/document/doi/10.1515/jqas-2014-0098/html

Demiralp, B., Colburn, C., & Koch, J. V. (2012). The effect of age, experience and managers upon baseball performance. *Journal of Economic Finance, 36*, 481-498. DOI 10.1007/s12197-010-9141-z

FanGraphs (2025). https://fangraphs.com/

Google. (2025). Google Colaboratory. Retrieved from https://colab.research.google.com/

Miceli, N. S. & Huber, A. D. (2009). 'If the team doesn't win, nobody wins:' A team-level analysis of pay and performance relationships in Major League Baseball. Journal of Quantitative Analysis in Sports, 5(2). https://doi.org/10.2202/1559-0410.1170

Spotrac (2025). https://spotrac.com/

Appendix A

Team	count	mean	std	min	25%	50%	75%	max
Arizona Diamond	10	111438511.9	3.43E+07	70762833	8.97E+07	103010585.5	1.29E+08	174000000
Atlanta Braves	10	149761568.9	4.79E+07	87622648	1.14E+08	139229717.5	1.93E+08	225315000
Baltimore Orioles	10	99782578	4.50E+07	30221166	6.87E+07	102320632.5	1.36E+08	163676616
Boston Red Sox	10	191850282.3	2.11E+07	162324847	1.81E+08	185703601	1.99E+08	235650000
Chicago Cubs	10	165937228.7	3.60E+07	116654522	1.35E+08	176152440	1.84E+08	216330000
Chicago White S	10	117058454.8	4.00E+07	59000000	9.88E+07	116506005.5	1.24E+08	181660734
Cincinnati Reds	10	106471413.8	1.66E+07	86374500	9.52E+07	102095000	1.17E+08	140000000
Cleveland Guard	10	85389033.5	2.88E+07	42310000	7.68E+07	86215121	8.91E+07	134350000
Colorado Rockie	10	127535945.5	2.34E+07	98261171	1.06E+08	130207868.5	1.40E+08	166250466
Detroit Tigers	10	133561763.3	3.79E+07	80398600	1.08E+08	130460000	1.62E+08	199750600
Houston Astros	10	155550727.3	5.37E+07	69064200	1.33E+08	167478799.5	1.79E+08	237303141
Kansas City Roy	10	106831239.5	2.18E+07	74110000	9.21E+07	112957262.5	1.15E+08	140925250
Los Angeles Ang	10	170953028.1	1.79E+07	146449583	1.59E+08	172206547	1.77E+08	204088094
Los Angeles Doc	10	241907599.2	3.85E+07	186140000	2.21E+08	229382639	2.65E+08	321000000
Miami Marlins	10	79384760	2.13E+07	47000000	6.86E+07	82392500	9.51E+07	111881100
Milwaukee Brew	10	100696728.5	2.14E+07	63061300	9.12E+07	96583997.5	1.08E+08	143000000
Minnesota Twins	10	122121563.8	1.31E+07	108077500	1.09E+08	119945212	1.34E+08	141271190
New York Mets	10	205513856.2	8.80E+07	99626453	1.55E+08	170110500	2.90E+08	334233332
New York Yanke	10	228705966.7	3.95E+07	168540000	2.05E+08	218695026.5	2.62E+08	293666666
Oakland Athletic	10	66184750	2.03E+07	32548334	4.97E+07	71572500	8.14E+07	92000000
Philadelphia Phil	10	178745851.8	6.07E+07	96850000	1.34E+08	171004500	2.30E+08	278000000
Pittsburgh Pirate	10	72220177.8	2.15E+07	35905000	6.36E+07	76500000	8.74E+07	100575946
San Diego Padre	10	145951209.1	5.58E+07	71624400	1.03E+08	133330576.5	1.91E+08	236962024
San Francisco G	10	170157297.3	2.34E+07	127889900	1.60E+08	172513361	1.85E+08	208510000
Seattle Mariners	10	126905741.4	3.00E+07	64553500	1.21E+08	129486190	1.51E+08	162480000
St. Louis Cardina	10	145450071.5	2.05E+07	120301957	1.25E+08	149258750	1.59E+08	180000000
Tampa Bay Rays	10	74485915.9	8.40E+06	60388600	7.07E+07	74498698.5	7.81E+07	88066012
Texas Rangers	10	156672055.2	4.04E+07	84868750	1.37E+08	144153686.5	1.82E+08	221955000
Toronto Blue Jay	10	169174939.5	4.00E+07	122506600	1.37E+08	165193452.5	1.99E+08	233000000
Washington Nati	10	141625354.7	3.95E+07	79983095	1.04E+08	163414016.5	1.67E+08	184000000

Table 1. Descriptive Statistics for Total Payroll

Team	count	mean	std	min	25%	50%	75%	max
Arizona Diamond	10	32.673	9.530938	13.15	28.42	32.56	36.0875	46.67
Atlanta Braves	10	36.397	12.671182	19.9	25.0725	38.81	41.7225	57.66
Baltimore Orioles	10	28.442	13.291572	11.89	14.68	32.865	38.8075	44.63
Boston Red Sox	10	39.897	8.572053	27.91	32.03	42.9	44.5175	51.27
Chicago Cubs	10	39.784	9.454868	22.65	36.99	41.525	44.2925	53.72
Chicago White S	10	25.993	12.387326	12.49	20.005	23.47	30.4925	55.87
Cincinnati Reds	10	26.93	8.468226	11.77	25.8475	27.215	31.635	39.65
Cleveland Guard	10	40.187	10.00829	28.49	31.16	40.44	44.32	56.51
Colorado Rockie	10	20.118	10.456344	1.08	14.25	21.64	26.8425	33.44
Detroit Tigers	10	22.477	9.845179	10.29	14.325	22.33	27.605	38.45
Houston Astros	10	50.166	9.845965	38.81	42.6475	47.65	56.8775	67.44
Kansas City Roy	10	25.136	9.836322	7.77	19.3325	25.205	30.8875	41.04
Los Angeles Ang	10	25.041	5.565354	14.58	23.22	27.205	28.2275	32.49
Los Angeles Doc	10	52.337	6.009189	42.72	48.825	50.645	56.09	62.44
Miami Marlins	10	35.473	9.233221	25.68	28.2875	33.55	39.88	55.59
Milwaukee Brew	10	37.05	7.227927	22.52	34.755	39.365	41.1675	46.83
Minnesota Twins	10	35.473	9.233221	25.68	28.2875	33.55	39.88	55.59
New York Mets	10	39.25	7.334521	29.15	34.765	39.405	43.225	52.58
New York Yanke	10	44.935	9.144774	28.26	39.3	48.57	52.0125	55.1
Oakland Athletic	10	26.497	12.823049	7.36	20.44	25.685	36.065	43.94
Philadelphia Phil	10	32.399	9.367259	19.08	25.5575	32.755	39.5575	47.6
Pittsburgh Pirate	10	24.163	7.772013	12.42	20.55	23.1	29.6775	36.22
San Diego Padre	10	31.643	10.025603	17.16	24.3175	31.215	39.0225	45.91
San Francisco G	10	32.554	4.151886	25.45	30.1575	32.055	35.755	38.22
Seattle Mariners	10	34.787	7.920866	22.74	27.7925	37.52	40.015	45.41
St. Louis Cardina	10	37.436	5.396055	30.41	32.5825	37.13	41.7475	45.51
Tampa Bay Rays	10	38.722	9.258677	28.05	31.8125	35.145	47.67	52.92
Texas Rangers	10	28.29	8.614051	17.69	23.5625	27.675	30.5475	48.87
Toronto Blue Jay	10	37.489	10.584155	21.87	27.0175	41.845	44.225	50.41
Washington Nati	10	31.836	15.330504	8.05	17.7225	34.685	44.9325	50.08

Table 2. Descriptive Statistics for Wins Above Replacement (WAR)

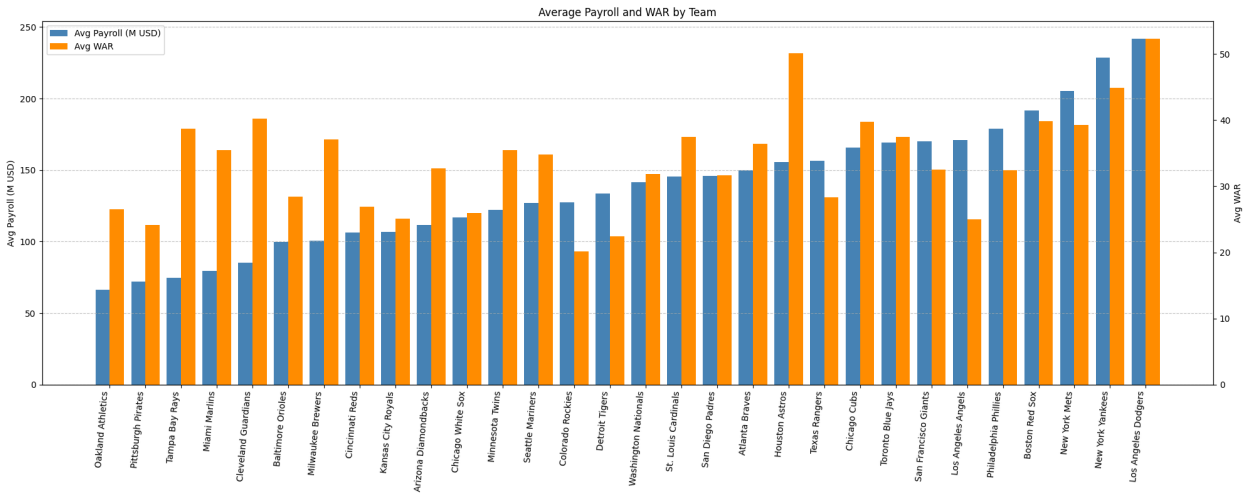

Figure 1. Relationship between Average Payroll and WAR across 10-year timeframe

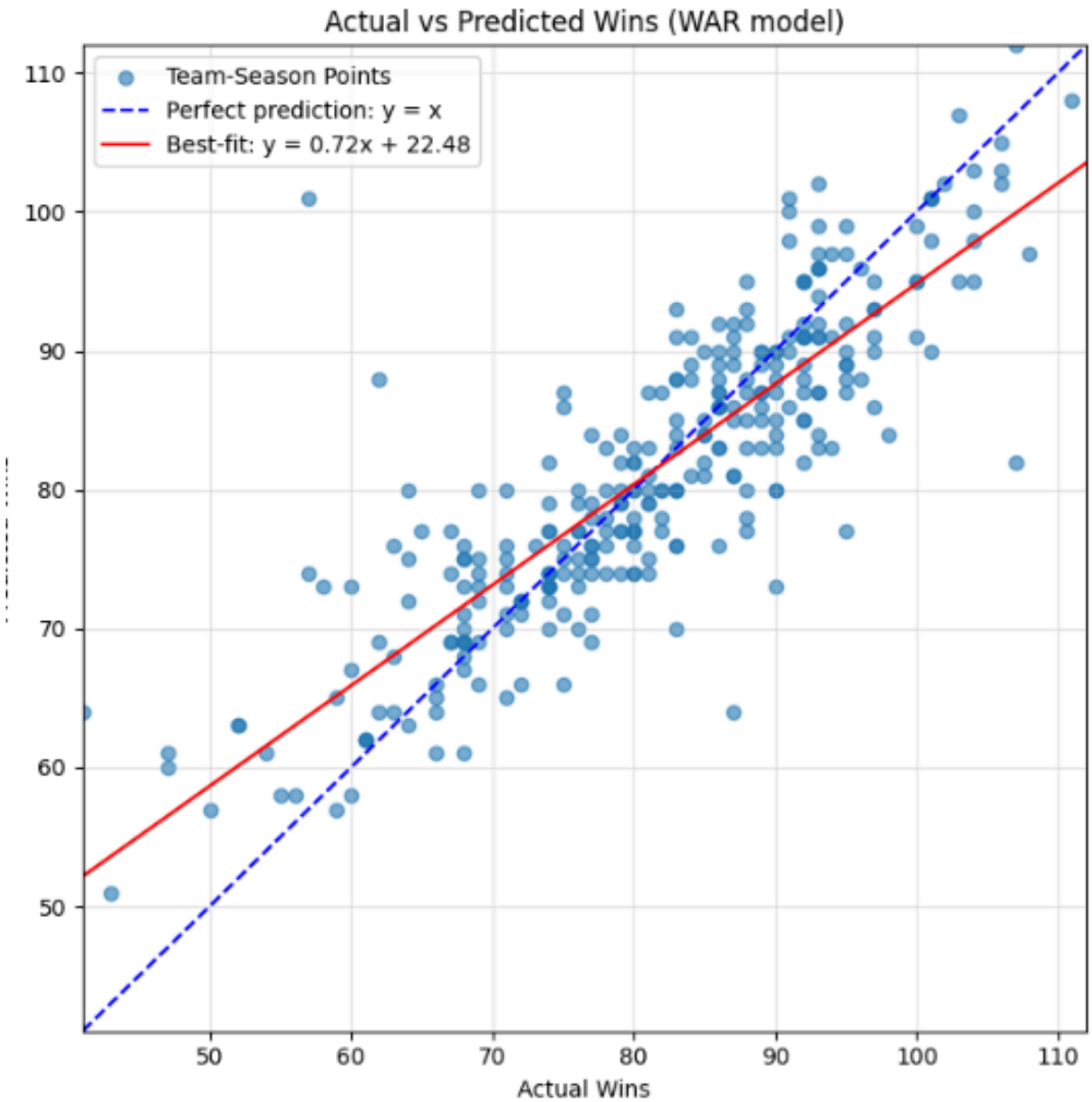

Figure 2. Scatter Plot of Predicted Wins against Actual Wins